The Rank the Task Demands: A Causal Rank Law for Matrix Memories Trained on Group Composition

Editors: Under Review for the NeurReps Workshop

Abstract

Matrix-valued memories make rank the natural budget of a learned representation: the number of independent directions a state spans bounds what it can bind, compose, and track. We report causal evidence, from two synthetic testbeds trained under a hard single-state bottleneck with an exact-continuous-recovery readout, that gradient descent recruits precisely the rank the task's algebra demands. On K-pair associative binding, where exact recovery provably requires state rank at least K, trained states develop effective rank tracking K (8.20 at K=8, 15.08 at K=16 for d=16), and train-time rank caps confirm the bound causally: at d=8, K=4, rank 3 yields zero exact recovery while rank 4 yields 0.94. On group-composition state tracking over five finite groups spanning the solvable/non-solvable divide, the recruited rank equals the group's minimal faithful real representation dimension $d_{\min}$ (Spearman $\rho = 0.9747$, the design's tie-capped maximum), the dimension-matched solvable/non-solvable pair S_4/A_5 is statistically equivalent under a pre-registered test, and a pre-registered force-rank razor is exact in both directions: one rank below $d_{\min}$, recovery is capped by the target's tied unit spectrum at $\sqrt{(d_{\min}-1)/d_{\min}} \leq 0.894$, below the 0.9 threshold in every group by construction, with observed cells at 76–95% (mean 88%) of that ceiling; at $d_{\min}$, not guaranteed a priori, recovery clears the anchor-relative bar in all five groups. Representation theory, not computational complexity class, sets what gradient descent buys.

Keywords: effective rank, fast weights, group representations, state tracking

1. Introduction

A matrix-valued memory represents by spanning: whatever a $d \times d$ state binds or tracks, it holds in the geometry of its column space, and rank is the budget it spends. Prior work treats this budget descriptively (Nazari and Rusch, 2026; Sun et al., 2026) or through hand-built constructions (Nichani et al., 2025); neither answers what a geometric account of learned representations needs answered: when a task's algebra fixes a minimal representational dimension, does gradient descent recruit exactly that dimension, and is it causally load-bearing?

We answer both halves affirmatively under one discipline: a hard single-state bottleneck (the decoder reads only one matrix Z, verified by a gradient blank-out test) and exact-continuous-recovery scoring (cosine against the true continuous target, never argmax over a codebook, which would let a rank-1 state recover on the order of d associations (Nichani et al., 2025)). Section 2 includes the provable K-pair-binding foundation. The headline is group-composition state tracking, where representation theory supplies the ground truth: across five finite groups spanning the solvable/non-solvable divide, the minimal faithful real representation dimension $d_{\min}$ predicts the recruited rank almost perfectly, a matched-dimension solvable/non-solvable pair lands statistically equivalent (Section 3), and a pre-registered force-rank razor is exact in both directions (Section 4): $d_{\min}-1$ is analytically incapable of exact recovery under the 0.9 threshold ($\sqrt{(d_{\min}-1)/d_{\min}} \leq 0.894$), and $d_{\min}$ suffices, which is not guaranteed a priori. Correcting the target's padding (Appendix C) recovers the razor's sufficiency result.

2. Tasks, Models, Instrument, and the Binding Foundation

Tasks. In the binding task, each episode presents K key–value pairs, the encoder writes one state $Z \in \mathbb{R}^{d\times d}$, prediction is the literal unbind Zk_j, and rec@0.9 is the fraction of queries with cosine to the true value above 0.9. The group task is the word problem of a finite group G: a word $w = g_{i_1}\cdots g_{i_L}$, drawn as a random walk on G's Cayley graph ($L \sim \mathrm{U}\{1,\ldots,8\}$), must map to $\rho_G(g_{i_1}\cdots g_{i_L})$ under a pinned orthogonal reference representation of dimension $d_{\min}(G)$, the minimal faithful real representation dimension. The groups are S_3, S_4, A_5, S_5, A_6 with $d_{\min} = 2, 3, 3, 4, 5$ (Appendix A); the first two are solvable, the rest are not, and non-solvable word problems are NC^1-complete (Barrington, 1989; Barrington and Thérien, 1988). $d_{\min}$ is a different, representation-theoretic axis; S_4/A_5, matched at $d_{\min} = 3$ with opposite solvability, are the designed dissociation pair.

Model. A Transformer encoder with d_{state} learned row-reader queries maps each word to a single state $Z(w) \in \mathbb{R}^{d_{\text{state}}\times d_{\text{state}}}$, $d_{\text{state}} = d_{\min} + 2$, leaving spare dimensions so over-recruitment is expressible. The target embeds the reference block-diagonally ($\rho_G(\cdot) \oplus I_2$ in the observational arm, $\rho_G(\cdot) \oplus 0$ in the causal-razor arm; Appendix C); the loss is cosine distance, the readout is fixed with no learned weights that could launder rank, and the decoder reads only Z (blank-out verified). Per-group step budgets (8k–40k, recorded in the design document's pre-registration) were pinned by convergence bars before any decisional cell ran.

Instrument. The learned state is defined only up to scale and an orthogonal change of basis (a Schur intertwiner): $Z(w) \approx c\,Q\,[\rho_G(w) \oplus \cdot\,]\,Q^\top$. We estimate the model's own dominant $d_{\min}$-subspace U from the SVD of the *centered* covariance of $Z(w)$ over held-out words, measure **restricted effective rank** (entropy effective rank of $U^\top Z(w) U$), and score **degauged recovery**: cosine after fitting (c, Q) on a fitting split, evaluated on a disjoint split, reported as rec@0.9. Centering is load-bearing (Appendix B); rank is invariant under the fitted gauge, so a rank-deficient state cannot be degauged into a full-rank one. For force-rank cells the decisional readout was pinned in advance to the conservative full-Q Procrustes variant (*crosscheck* rec@0.9).

The provable foundation (binding).

Fact 1 *If $Zk_j = v_j$ exactly for K bindings with linearly independent keys and values, then $ZK_{\text{mat}} = V_{\text{mat}}$, so $K = \mathrm{rank}(V_{\text{mat}}) \leq \mathrm{rank}(Z)$.*

The bound is classical (Kohonen, 1972; Anderson, 1972), holding only under exact continuous recovery. Gradient descent recruits the rank: at $d = 16$ effective rank reaches 8.20 at $K = 8$ and 15.08 at $K = 16$, and the recruited rank is causally necessary: a spectral rank cap at $d = 8$, $K = 4$ leaves exact recovery ≤ 0.0004 for every cap $k \leq 3$ and restores it to 0.940 at $k = 4$, a step exactly at the provable bound; the same operator composes exactly (Z^h, 21 self-applications) in four of five seeds, a composition-stability probe periodicity-equivalent to depth 5 under the single 8-cycle target (Liu et al., 2023) (Appendix E).

3. The Rank Law on Group Composition, Observed

Figure 1 (left) gives the observational leg: per-group mean restricted effective rank lands within 4.9–10.2% of $d_{\min}$ at every group, and all 19 seeds sit inside the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band (means and per-seed values in Appendix B).

Spearman correlation between recruited rank and $d_{\min}$ is $\rho = 0.9747$, the maximum this design can express, since the S_4/A_5 tie caps ρ below 1; under the exact permutation null, $P(\rho \geq 0.8) = 8/120 \approx 6.7\%$ (0.8 is pre-registered, below the next achievable level 0.87), so by pre-registration this leg is corroborating rather than independently decisive (a disclosed length-robustness split is in Appendix B).

Dimension, not solvability. If recruited rank were set by computational class, the marquee pair should separate; if by representation dimension, coincide. A pre-registered Welch two-one-sided equivalence test on restricted effective rank (margin ± 0.5 rank-units; $n = 5$ per group) observes a difference of $+0.019$ rank-units (se 0.037, df 7.8), with both one-sided tests passing at roughly seven times the critical value ($t = 13.06$ and 14.12 against $t_{\text{crit}} = 1.865$): equivalence is declared decisively. The pair lands together on dimension, where the rank law puts it.

4. The Causal Razor

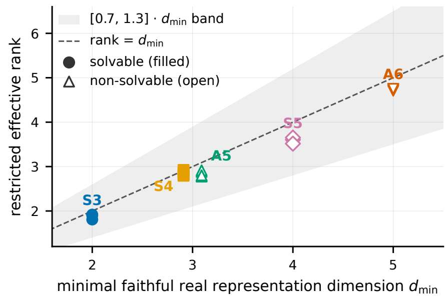
The decisive test intervenes on rank: the encoder's output is spectrally truncated to rank $k \in \{d_{\min}-1, d_{\min}, d_{\min}+1\}$ throughout training, against the zero-padded target $\rho_G(\cdot) \oplus 0$ of rank exactly $d_{\min}$, alongside an unconstrained anchor from the same family. Pre-registered reading: below $d_{\min}$, a sound readout pins recovery to zero by geometry, making that arm an integrity control whose registered trigger (any recovery there) would indicate an instrument leak; the live prediction is that $k \geq d_{\min}$ recovers past 0.9× the anchor's crosscheck rec@0.9.

Figure 1 (right) and Figure 2 show the result on the pre-pinned crosscheck readout. Necessity below $d_{\min}$ is an analytically forced floor: against a $d_{\min}$-tied-unit-singular-value target, the maximum cosine any rank-$(d_{\min}-1)$ operator can reach is $\sqrt{(d_{\min}-1)/d_{\min}} \leq 0.894$, below the 0.9 threshold in every group by construction, so rec@0.9 = 0.000 at $k = d_{\min}-1$ follows regardless of training quality; observed cells still reach 76–95% of that ceiling (mean 88%; Appendix B), evidence of near-optimal training under the cap. The zero is noiseless in all five groups and all four S_3 seeds. Sufficiency at $d_{\min}$ carries the causal claim, since no such bound guarantees a rank-exactly-$d_{\min}$ solution is reachable: S_4, A_5, S_5, A_6 clear their bars outright, and S_3's decisive cell, inside the pre-stated ± 0.05 marginality trigger, was extended to four seeds: seed-mean 0.5625 against the fixed 0.495 bar confirms the fifth group (per-seed detail in Appendix B). Both marquee groups confirm at $d_{\min}$, meeting the causal criterion.

Identity-padding the target confounds the razor by handing every capped arm a free rank-2 loss reduction (Appendix C); the zero-padded target above removes the confound.

5. Related Work and Limitations

Chughtai et al. (2023) reverse-engineers group multiplication via representation theory; here, rank itself, not an irrep decomposition, is the measured and manipulated quantity.



G	anchor	k=$d_{\min}$−1	k=$d_{\min}$	k=$d_{\min}$+1	bar
S_3	0.550	0.000	0.450*	0.550	0.495
S_4	0.650	0.000	**0.800**	0.950	0.585
A_5	0.700	0.000	**0.700**	0.750	0.630
S_5	0.500	0.000	**0.600**	0.550	0.450
A_6	0.650	0.000	**0.650**	0.700	0.585

Figure 1: Left: recruited rank follows representation dimension, not solvability. Each point is one seed's restricted effective rank vs. its group's $d_{\min}$ (filled: solvable; open: non-solvable); all 19 seeds are in band, and S_4/A_5 coincide at $d_{\min} = 3$. Right: the causal razor on the zero-padded rank-$d_{\min}$ target (crosscheck rec@0.9 by force-rank k, one seed per cell, unconstrained anchor, pre-registered 0.9×anchor bar): exactly 0.000 one rank below $d_{\min}$ everywhere (the forced ceiling is ≤ 0.894), clears the 0.9×anchor bar at $d_{\min}$ in all five groups (bold; S_3 via seed-mean, asterisked). *extended to four seeds under the pre-stated ±0.05 trigger: seed-mean 0.5625 clears the fixed 0.495 bar.

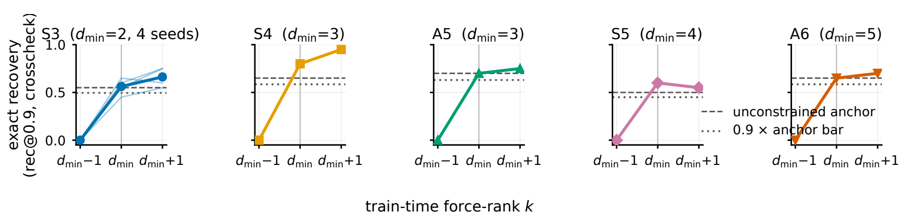


Figure 2: Exact recovery is a step function at $d_{\min}$: per group, crosscheck rec@0.9 on held-out words at force-rank $k \in \{d_{\min}-1, d_{\min}, d_{\min}+1\}$, vs. the unconstrained anchor (dashed) and the 0.9×anchor bar (dotted). S_3 overlays its four seeds (thin lines; bold mean); its below-$d_{\min}$ zero is unanimous.

Nazari and Rusch (2026) and Sun et al. (2026) measure effective-rank dynamics in pretrained linear-attention models, observationally, with no task algebra fixing a required dimension; Mishra et al. (2026) train matrix-state RNNs on S_3 composition without a rank intervention. Grazzi et al. (2025), Siems et al. (2025), and Merrill et al. (2024) characterize which word problems recurrent architectures can express, and Delétang et al. (2023) benchmark sequence architectures across the formal-language hierarchy; the marquee equivalence instead sorts by representation dimension.

Limitations. All results are on sub-1M-parameter synthetic testbeds, not claims about pretrained language models. Razor cells are single-seed except S_3 (four seeds); S_4/S_5's anchor-bar margins and S_3/S_5's soft convergence at the shortest pinned budget (Appendices B, D) both read directionally, not as an unqualified 5/5.

Appendix A. The Five Groups and Reference Representations

G	$\lvert G\rvert$	$d_{\min}$	solvable	realization of ρ_G
S_3	6	2	yes	standard (dihedral)
S_4	24	3	yes	cube rotations
A_5	60	3	no	icosahedral rotations
S_5	120	4	no	standard (zero-sum)
A_6	360	5	no	standard (zero-sum)

Table 1: The five task groups. $d_{\min}$ is the minimal faithful real representation dimension; each reference representation is orthogonal by construction and verified faithful (injective on G) at build time. S_4/A_5 form the marquee pair: matched $d_{\min}$, opposite solvability.

Appendix B. Per-Seed Observational Rank

G	$d_{\min}$	n	restricted eff. rank	band	in band
S_3	2	3	1.877 ± 0.060	$[1.4, 2.6]$	3/3
S_4	3	5	2.852 ± 0.054	$[2.1, 3.9]$	5/5
A_5	3	5	2.832 ± 0.062	$[2.1, 3.9]$	5/5
S_5	4	3	3.591 ± 0.069	$[2.8, 5.2]$	3/3
A_6	5	3	4.736 ± 0.023	$[3.5, 6.5]$	3/3

Table 2: Restricted effective rank of the unconstrained trained state (mean $\pm$ sd over seeds) against each group's $d_{\min}$, with the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band. Every seed of every group is in band; Spearman $\rho = 0.9747$ is the tie-capped maximum; the dimension-matched pair S_4/A_5 differs by 0.019 rank-units.

As a fraction of the geometrically forced ceiling $\sqrt{(d_{\min}-1)/d_{\min}}$, the below-$d_{\min}$ cross-check cosines read S_3 86.3%, S_4 91.2%, A_5 94.9%, S_5 75.7%, and A_6 93.5% (mean 88.3%). S_5 is the outlier, consistent with its soft-convergence flag (Appendix D).

Per-group deviation of the mean from $d_{\min}$ (Table 2): S_3 6.1%, S_4 4.9%, A_5 5.6%, S_5 10.2%, A_6 5.3%.

S_4 and S_5's razor $k{=}d_{\min}$ cells (Figure 1, main text) numerically exceed their own unconstrained anchor by 0.10–0.15; plausibly the capped arm skips the ambient dimensions the zero-padded anchor must still learn to null under the same fixed step budget, an unconfirmed hypothesis.

The instrument's centering step is load-bearing: an uncentered lens scores a flawless synthetic model 0.705 where the centered production lens scores 0.9996 on identical data. The disclosed length-robustness split referenced in Section 3 scores only words of length $L \geq 2$ (the $L = 1$ read is attention-degenerate and was demoted with a mechanism note in the run records): it moves per-group mean recovery cosine by at most 0.013 (per-seed at most 0.041), with no group-selective divergence. Per-seed, S_3's $k{=}d_{\min}$ cell clears its own seed's 0.9×anchor bar in 2 of 4 seeds (seed 1: +0.010; seed 3: +0.110; seed 0: −0.045; seed 2: −0.120); the self-referential bar (0.9× the seed-mean anchor, 0.574) narrowly fails the seed-mean $k{=}d_{\min}$ value (0.5625), which is why the main text uses the fixed pre-registered literal (0.495) rather than a self-referential recompute.

Appendix C. The Ambient-Identity Tax in Detail

The identity-padded force-rank target $\rho_G(\cdot) \oplus I_2$ has rank $d_{\text{state}} = d_{\min} + 2$ for every group. Because the identity block is constant across words, it is the cheapest reliable loss reduction available, and a rank-capped arm buys it before the group representation; the residual budget for ρ_G is $k - 2$, below $d_{\min}$ at every grid point, so no capped cell under identity-padding ever tested the law's confirm direction. Three raw-artifact signatures established this: (i) force-rank cells' direct cosine matched the rank-k optimum $\sqrt{k/d_{\text{state}}}$ with mean absolute deviation 0.028 across all 39 cells (maximum 0.166; the two S_5 below-$d_{\min}$ cells are the only outliers, an additional disclosed optimization shortfall), so the arms trained to their rank-constrained ceiling rather than under-training; (ii) centered restricted rank of capped arms landed near $k - 2$, the residual budget; (iii) in the zero-padded grid, a deliberately eye-padded corroboration arm reproduces the failure on demand at raw $k = d_{\min}{+}1$ (effective budget $d_{\min}{-}1$) while the tax-paid point recovers. The zero-padded grid's 30 cells were configuration-verified one-by-one against a manifest re-derived independently from the design record (steps, padding mode, and force-rank value per cell; zero skipped steps).

Appendix D. Soft Convergence at the Shortest Pinned Budgets

G	anchor	$k{=}d_{\min}{-}1$	$k{=}d_{\min}$	$k{=}d_{\min}{+}1$	bar (0.92)
S_3	0.914	0.665	0.900	0.903	fail (all 4)
S_5	0.876	0.801	0.879	0.877	fail (all 4)

Table 3: Convergence health (gate1a, min validation cosine over $L \in [2, 5]$) for every S_3/S_5 razor cell at the shortest pinned step budget. All eight cells fall short of the 0.92 margin; S_5's 0.876–0.879 range is a larger shortfall than S_3's 0.900–0.914. The necessity leg (Section 4) is unaffected by construction; the sufficiency leg for these two groups should be read as directionally consistent under disclosed soft convergence.

Appendix E. Depth-21 Periodicity Under the Single K-Cycle

The binding task's held-out composition probe (Section 2) uses a single Hamiltonian K-cycle, chosen because a general permutation decomposes into short disjoint cycles that collapse held-out hops into trivial or in-distribution queries. Under a single 8-cycle, $\pi^{21} = \pi^{21 \bmod 8} = \pi^5$ exactly, so nominal depth 21 shares its target with the already-tested depth

5; the two are not distinct group-theoretic targets. What depth 21 measures instead is numerical stability under 21 sequential self-applications of the trained operator: in the four converged seeds, rec@0.9 at $h = 5$ and $h = 21$ both read ≈ 1.0, a non-trivial statement that repeated self-application does not amplify error; the fifth (still-transitioning) seed reads 0.303 at $h = 5$ and 0.0001 at $h = 21$, the same underlying imperfection compounding under repetition.

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